Reading report

PhD candidate: Maxime Deforche
Title of the doctoral dissertation: Management and Analysis of Short Texts: A Graph-Based Approach

Member of the examination board: Kristoffel Demoen

A preliminary caveat: this reviewer is not at all an expert in the field of computer science (in which the PhD is being awarded), but in that of literary studies (moreover with a focus on qualitative, not quantitative analysis). As the director of the Database of Byzantine Book Epigrams (DBBE), however, I am thoroughly familiar with the textual data used as the test case in this project (the 'short texts' referred to in the title). I will therefore primarily assess the accuracy and relevance of the way in which the textual data has been handled in this thesis, and I have read the work partly with a 'what's in it for us' attitude.

I must add that, contrary to my expectations, I was generally able to follow the line of argumentation and its exposition (albeit not down to the technical details of calculations, formulae and the like, of course). Whether this is a compliment to the PhD candidate or not, I leave open to interpretation. In any case, it is intended as praise. My report further follows the structure from the template.

1. Content and scientific level

Naturally, I am not in a position to make a substantive assessment of the scientific merits of this work in the field of computer science. What I have, however, noted with great appreciation is that the thesis definitely meets the conceptual criteria listed as guiding questions in the report template:

- it contains a clear and well-motivated statement of the research questions and objectives (I found the introductory chapter 1 excellent);
- the candidate's own contributions are clearly described and positioned within the state of the art, which is summarized in each chapter under the heading of 'Related Work';
- the methodology is duly motivated and described in an almost didactic manner;
- the research data appear to be rigorously processed. The qualification "appear to be" is due to two considerations (which are no points of criticism): [1] the dissertation gives a necessarily partial description of the research data (including the data selection and [pre]processing) and the tools that have been produced to analyze them – the graph database itself remains a kind of mystery, to me at least; [2] it is unclear to what extent Maxime is responsible for the basic decisions about the selection and [pre]processing of the Greek textual data;
- the results of the experiments are clearly presented and critically discussed;
- whether the scientific results extend the state of the art in computer science engineering I cannot tell; I can confirm that they have the potential to have an impact on academic research in the field of automated analysis of historical texts in "low-resource languages".

I was especially intrigued and charmed by research questions 3 and 4, and by the chapters in which these RQs are addressed in most detail, i.e. chapters 4 to 6. The final chapter really opens promising practical applications when it comes to the (desirable) revision of partitions of the DBBE corpus into

different (numbers of) clusters, and possibly also the way we cluster them (nested partitions, dendrograms). Wish we had this know how at the start, not at (nearly) the end of our project.

2. Formal aspects

I was favorably impressed by the quality and the clarity of the academic language used in this dissertation, as well as by its solid and coherent overall organization. This is indeed a unified thesis, despite the fact that the core chapters are based on separate publications.

Chapters 2-6 build on each other in a convincing manner. They all present a similar structure, starting with a helpful introduction, a systematic return to the overarching research questions, and transparent references to the publications on which they are based. The reader is given a correct indication of the novelty of each new step and a positioning within related work. The description of the concepts and the experiment(s) is generally clear (even if at times way too technical for me) and supported with many relevant graphs and tables. Each chapter closes with excellent summaries and conclusions, including intelligent and self-critical discussions of the limitations of the research methodologies and results. These discussions lead fluidly to the next chapter, with eloquent and logical transitions.

3. Critical remarks/questions and potential topics for the discussions

Preprocessing and similarity scores (chapters 2 and 3). Here's the Hellenist speaking. The preprocessing has tackled the phenomenon of itacism (p. 47, correctly explained in section 1.3), but not other vowel and consonant shifts that equally may explain spelling differences. To stick with an example taken from tables 2.1 and 2.2: the homophonous ξένη and ξένοι are collapsed into ξένι, hence forming one word node, whereas the equally homophonous θαλαττεύοντες and θαλατέβονταις remain unchanged, despite the three spelling differences within one word (compare the grid in figure 3.3, p. 74). My question to Maxime would obviously not be to give a linguistic justification of this diverse treatment of similar phonological phenomena, but to give an indication of its mathematical impact on all further calculations. Extending preprocessing with ο = ω, for instance, would, for sure, change table 2.3 p. 49, since the most frequent verses 4 and 5 are phonologically identical and should therefore be added up and occupy rank 2.

Types and occurrences. Figure 3.9 visualizes groups of epigrams belonging to type 2148, and two of these epigrams (occurrences, in principle) are discussed in more detail p. 96. One of these, 21182, is in fact not a real occurrence (see its editorial status: we have not seen the manuscript), but provisionally duplicates the text of the type. Again, no worries, but what intrigues me is that this poem, which is identical to the type, finds itself at such a marginal position in the visualization on p. 95. How to explain this? (I have an idea but would wonder what Maxime can make out of it.) By the way, the visualizations in this chapter are really impressive, and I applaud the capacity to trace results back to their constituent scores (wish I could do that myself).

Very small remark (but with some resonance at UGent): the motto of chapter 4 is indeed often ascribed to Aristotle, but it not an authentic quotation; in his *Metaphysics* VIII.6 (1045a), Aristotle says "In all things which have a plurality of parts, and which are not a total aggregate but a whole of some sort distinct from the parts, there is some cause." (This is a kind of veracity issue.)

Metadata. In chapter 4, five metadata fields are considered for the experiments: geographical location, genre, metre, scribe, subject. Why not date?

Ranking techniques for context-aware similarity-confidence pairs. Fascinating and relevant, and as far as I could follow theoretically convincing. Yet, one line in table 5.1 puzzles me: rank #4, epigram 20915, has exactly the same text as the reference epigram 17754 (including the spelling error ευρι**ππ**ίδου absent from #1-3) and yet receives a similarity score of only 0.750 (#1-3 have 1.000), which is closer to that of #5 in both columns (0.699 and 0.575 respectively), whereas these poems (22372 and 19670) are *completely* different. How to explain this counter-intuitive score?

Geographical querying. I didn't understand the meaning of the second use case of chapter 5.

Further items that I might bring up at the internal and/or the public defense:

- a retrospective reflection on the doctoral process and the collaboration within the GOA team
- the balance between tool development and knowledge expansion
- the groups of potential users of the graph database (for DBBE), related to distinct activities (management, curating, navigating, querying, analysis, exploration, interpretation, visualization… of 'our' textual data)
- possible other textual data (corpora) for similar/further research
- (some of) the mottoes chosen as epigraphs to the chapters
- the possibility of automated detection of 'outliers' that would warrant closer inspection: especially chapters 4-6 include several examples, for instance of epigrams that belong to a same expert-curated type but fail to connect to the main group in the visualization of the text nodes (one telling example is figure 4.5 p. 136; Maxime does not identify the two epigrams that remain unconnected to the main group here – my guess would be: 26591 and 34580).

4. Suggestions for modification of the doctoral dissertation

None (except perhaps for the pseudo-Aristotelian motto of chapter 4).